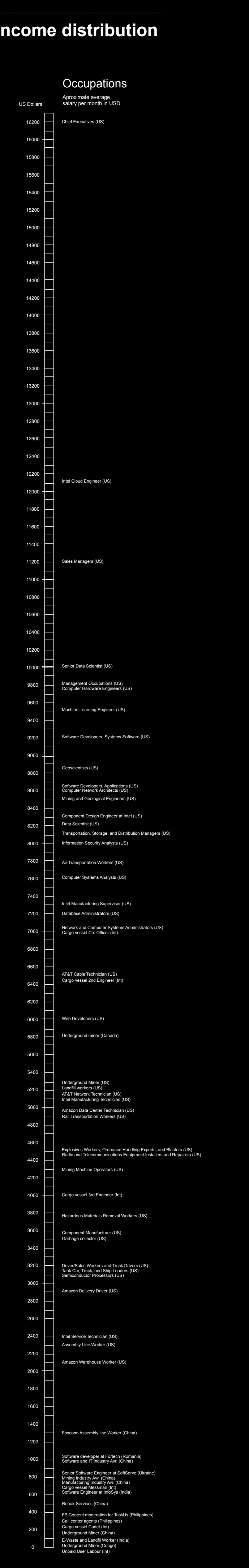
Anatomy of an AI system

An anatomical case study of the Amazon echo as a artificial intelligence system made of human labor

Income distribution

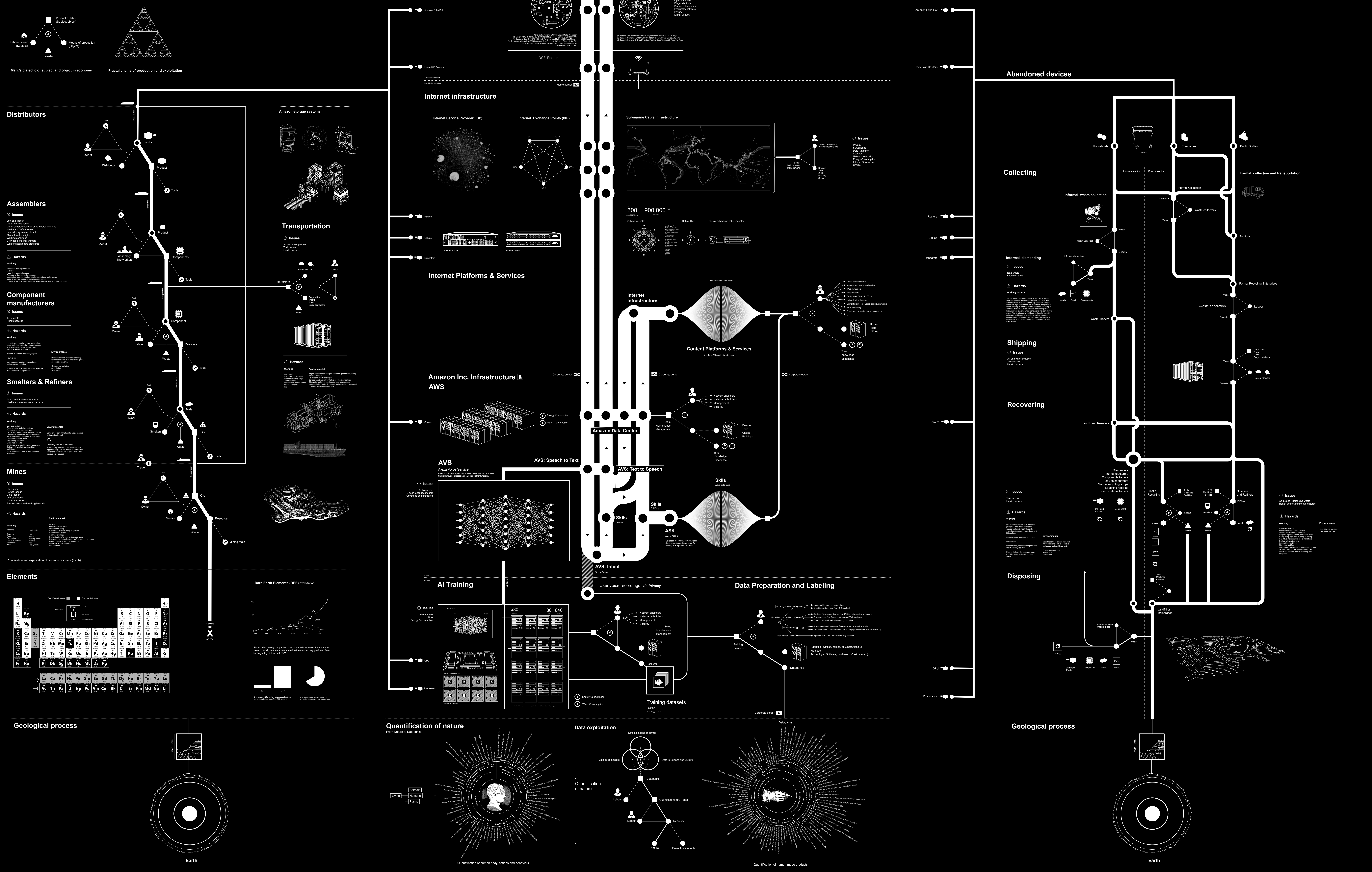


Occupations

Aproximate average salary per month in USD

US Dollars

Chief Executives (US)
Intel Cloud Engineer (US)
Sales Managers (US)
Senior Data Scientist (US)
Management Occupations (US)
Computer Hardware Engineers (US)
Machine Learning Engineer (US)
Software Developers, Systems Software (US)
Geoscientists (US)
Software Developers, Applications (US)
Computer Network Architects (US)
Mining and Geological Engineers (US)
Component Design Engineer at Intel (US)
Data Scientist (US)
Transportation, Storage, and Distribution Managers (US)
Information Security Analysts (US)
Air Transportation Workers (US)
Computer Systems Analysts (US)
Intel Manufacturing Supervisor (US)
Database Administrators (US)
Network and Computer Systems Administrators (US)
Cargo vessel Ch. Officer (Int)
AT&T Cable Technician (US)
Cargo vessel 2nd Engineer (Int)
Web Developers (US)
Underground miner (Canada)
Underground Miner (US)
Landfill workers (US)
AT&T Network Technician (US)
Intel Manufacturing Technician (US)
Amazon Data Center Technician (US)
Rail Transportation Workers (US)
Explosives Workers, Ordnance Handling Experts, and Blasters (US)
Radio and Telecommunications Equipment Installers and Repairers (US)
Mining Machine Operators (US)
Cargo vessel 3rd Engineer (Int)
Hazardous Materials Removal Workers (US)
Component Manufacturer (US)
Garbage collector (US)
Driver/Sales Workers and Truck Drivers (US)
Tank Car, Truck, and Ship Loaders (US)
Semiconductor Processors (US)
Amazon Delivery Driver (US)
Intel Service Technician (US)
Assembly Line Worker (US)
Amazon Warehouse Worker (US)
Foxconn Assembly line Worker (China)
Software developer at Fortech (Romania)
Software and IT Industry Avr. (China)
Senior Software Engineer at SoftServe (Ukraine)
Mining Industry Avr. (China)
Manufacturing Industry Avr. (China)
Cargo vessel Messman (Int)
Software Engineer at InfoSys (India)
Repair Services (China)
FB Content moderation for TaskUs (Philippines)
Call center agents (Philippines)
Cargo vessel Cadet (Int)
Underground Miner (China)
E-Waste and Landfill Worker (India)
Underground Miner (Congo)
Unpaid User Labour (Int)



Marx's dialectic of subject and object in economy

Fractal chains of production and exploitation

Distributors

Assemblers

Issues

Low paid labour
Illegal working hours,
Unfair compensation for unscheduled overtime
Health and Safety issues
Internship system exploitation
Migrant workers rights
Working conditions
Crowded dorms for workers
Workers health care programs

Hazards

Working

Hazardous working conditions
Explosions
Hazardous chemical exposure
Exposure to dust and toxic substances
Inconsistent health and safety policies, procedures and practices
Major depression and the risk of attempted suicide
Ergonomic hazards : body positions, repetitive work, shift work, and job stress

Component manufacturers

Issues

Toxic waste
Health hazards

Hazards

Working

Use of toxic materials such as arsine, phosphine and others potentially expose workers to health hazards which include cancer, miscarriages and birth defects.

Irritation of skin and respiratory organs

Neurotoxins

Low-frequency electronic magnetic and radiofrequency radiation

Ergonomic hazards : body positions, repetitive work, shift work, and job stress

Environmental

Use of hazardous chemicals including hydrochloric acid, toxic metals and gases, and volatile solvents.

Groundwater pollution
Air pollution
Toxic waste

Smelters & Refiners

Issues

Acidic and Radioactive waste
Health and environmental hazards

Hazards

Working

Low-level radiation
Airborne metal and silica particles
Contact with corrosive chemicals
Dangerous gases, vapors, fumes and dusts
Heavy lifting, high-force pushing or pulling,
Repetitive motion during use of hand tools
Contact with molten metal
Hot working conditions
Slips, trips and falls
Moving parts on machinery and equipment that can cut, crush, impale, or strike individuals
Noise and vibration due to machinery and equipment

Environmental

Large proportion of the harmful waste products toxic waste disposal

Refining rare earth elements

After refining one ton of rare earth elements, approximately 75 cubic meters of acidic waste water and about one ton of radioactive waste residue are produced.

Mines

Issues

Hard labour
Forced labour
Child labour
Low paid labour
Conflict minerals
Environmental and working hazards

Hazards

Working

Accidents

Cave-ins
Flood
Gas explosions
Chemical leakage
Electrocution
Fires

Health risks

Dust
Radon
Welding fumes
Mercury
Noise
Heavy loads

Environmental

Erosion
Formation of sinkholes
Loss of biodiversity
Devastation of surrounding vegetation
Contamination of soil
Acid rock drainage
Contamination of ground and surface water
High concentrations of arsenic, sulfuric acid, and mercury
Affecting health of the local population
Noise dust and visual pollution
Deforestation

Privatization and exploitation of common resource (Earth)

Elements

Amazon storage systems

Transportation

Issues

Air and water pollution
Toxic waste
Health hazards

Hazards

Working

Cargo Shift
Cargo falling from height
Dust from working cargo
Turbulent Seas
Maintenance related injuries
Mooring Hazards
Fire

Environmental

Air pollution (conventional pollutants and greenhouse gases)
Acoustic pollution
Devastating effects of oil spills
Sewage, wastewater from toilets and medical facilities,
Bilge water leaks from engine and machinery spaces
Impact of ballast water discharges on the marine environment.
Collisions with marine mammals,

Rare Earth Elements (REE) exploitation

'Since 1980, mining companies have produced four times the amount of many, if not all, rare metals compared to the amount they produced from the beginning of time until 1980.'

On average, a 21st century citizen uses ten times more minerals than one of the 20th century.

In a single Iphone there is about 75 elements : two-thirds of the periodic table.

Geological process

Human operator

Issues

Privacy
Digital security
Unpaid immaterial labour

Domestic infrastructure

Amazon **Echo Dot**

(1) Texas Instruments DM3725 Digital Media Processor
(2) Micron MT46H64M32LFBQ 256 MB (16 Meg x 32 x 4 Banks) LPDDR SDRAM
(3) Samsung KLM4G1FEPD 4GB High Performance eMMC NAND Flash Memory
(4) Qualcomm Atheros QCA6234 Integrated Dual-Band 2x2 802.11n + Bluetooth 4.0 SiP
(5) Texas Instruments TPS65910A1 Integrated Power Management IC
(6) Texas Instruments DAC

(1) National Semiconductor LP55231 Programmable 9-Output LED Driver (x4)
(2) Texas Instruments TLV320ADC3101 92dB SNR Low-Power Stereo ADC (x4)
(3) Texas Instruments SN74LVC74A Dual Positive-Edge-Triggered D-Type Flip-Flops

Issues

Right to repair
Open schematics
Diagnostic tools
Planned obsolescence
Proprietary software
Privacy
Digital Security

WiFi Router

Internet infrastructure

Issues

Privacy
Surveillance
Data Retention
Security
Network Neutrality
Energy Consumption
Internet Governance
Sharks

Internet Platforms & Services

Internet
Infrastructure
Servers and Infrastructure
Owners and investors
Management and administration
Web developers
Programmers
Designers (Web, UI, UX ..)
Network administrators
Content producers (users, editors, journalists)
PR & Marketing
Free Labour (user labour, volunteers ,,)
Devices
Tools
Offices
Time
Knowledge
Experience

Content Platforms & Services

(eg. Bing, Wikipedia, Weather.com ..)

Amazon Inc. Infrastructure
AWS

AVS

Alexa Voice Service

Alexa Voice Service performs speech to text and text to speech, Natural language processing (NLP) and other functions.

Issues

AI 'black box'
Bias in language models
Unverified and unaudited

ASK

Alexa Skill Kit

Collection if self-service APIs, tools, documentation and code used for making of 3rd party Alexa Skills

AI Training

Issues

AI Black Box
Bias
Energy Consumption

Each of 80 nodes communicate updates to the model to all other nodes every second

Training datasets

+20000

Hours of tagged content

Data Preparation and Labeling

Unrecognized labour
Immaterial labour (eg. user labour)
Unpaid crowdsourcing (eg. ReCaptcha)
Unpaid or Low paid labour
Students, Volunteers, Interns (eg. TED talks translation volunteers)
Crowdworkers (eg. Amazon Mechanical Turk workers)
Outsourced services in developing countries
Professionals
Science and engineering professionals (eg. research scientist)
Information and communications technology professionals (eg. developers)
Non-Human Labour
Algorithms or other machine learning systems
Training datasets
Facilities (Offices, homes, edu.institutions ..)
Methods
Technology (Software, hardware, infrastructure ..)
Databanks
Corporate border
Databanks

Quantification of nature

From Nature to Databanks

Quantification of human body, actions and behaviour

Data exploitation

Quantification of human-made products

Abandoned devices

Collecting

Informal dismantling

Issues

Toxic waste
Health hazards

Hazards

Working Hazards

The hazardous substances found in the e-waste include substantial quantities of lead, cadmium, chromium and flame-retardant plastics. Cathode ray tubes and components with high lead content are considered dangerous to health. Inhaling or handling such substances and being in contact with them on a regular basis can damage the brain, nervous system, lungs, kidneys and the reproductive system Working in poorly-ventilated enclosed areas without masks and technical expertise results in exposure to dangerous and slow-poisoning chemicals. Due to lack of awareness, workers are risking their health and environment as well.

Shipping

Issues

Air and water pollution
Toxic waste
Health hazards

Recovering

Issues

Toxic waste
Health hazards

Hazards

Working

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Noise and vibration due to machinery and equipment

Environmental

Harmful waste products
toxic waste disposal

Disposing

Geological process

Vladan Joler andKate Crawford (2018)